Experiment No.: 2

Experiment Title: Basics of microcontroller (Time dependent execution and communication)

Group No.: B6

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Results for the group task:

Task no.	Topic	Task	Report
1	Using timer millis()	Blink the built-in LED without using delay() function this time. Write a program to light up the built-in LED for 1 second and then turn it off for the next 3 seconds.	I changed the value of the variable “interval” to 3000 and “duration” to 1000.
2	Interrupts and ISR	Write a program to light up the built-in LED (and stay ON forever) when interrupt pin 2 detects a FALLING edge. Use pinMode(interruptPin, INPUT_PULLUP) to maintain the interrupt pin 2 at 5V always. Connect pin 2 to GND once to obtain a falling edge. Remove the GND connection to pin 2. Use the RESET button on Arduino to reset the microcontroller operation.	(a) Voltage output at pin 2 before connecting it to GND: 4.77V. (b) The voltage output at pin 2 after connecting it to GND once and then disconnecting it: 4.77V (c) LED turned ON after connecting it to GND and remained ON after disconnecting it. (d) When RESET button of Arduino is pressed, pin 2 voltage is 4.89V and LED get turned OFF.
3	Interrupts and ISR	The code for task 3 is very similar to task 2 code, except it introduces a ten-second delay in the main loop unlike task 2 code. Connect pin 2 to GND once to obtain a falling edge. Remove the GND connection to pin 2.	(a) After connecting pin 2 to GND once and then removing this connection, the LED turned OFF instantly. (b) No, there is no effect of changing the delay duration in the main loop, because on falling edge, interrupt will be triggered immediately as it works independently of the loop.

4	Serial communication: UART (transmitter)	The code for task 4 transmits an integer varying from 1 to 10 printing each consecutive number after a delay of certain seconds (identify this delay duration given in the provided code for Task 4). Change the provided code to produce each consecutive number on the serial monitor after a delay of 2 seconds from the previously produced number.	(a) A new number is produced in the serial monitor after every 6 seconds. (b) I changed the delay() to delay(2000); in the for loop to get a delay of 2 seconds.
5	Serial communication: UART (receiver)	Upload the code for Task 5 to receive an integer from Serial Monitor (use Serial.parseInt()). Send any number between 0-10 from the computer by typing it into the Serial monitor and pressing enter. Blink the built-in LED that number of times within the next 5 seconds after receiving the number.	I entered '10' on the serial monitor, and the LED started blinking 10 times i.e., number of glows: 10 and duration of each glow is 5000 / (2 * 10) = 250ms, and the total time for all the glows is 5 seconds.
6	Serial communication: UART (transmitter and receiver)	Upload the provided transmitting code ('MicrocontrollerA_serial_transmitter.ino') to board A and receiving code ('MicrocontrollerB_serial_receiver.ino') to board B. Do the connections. Using these codes, Board A transmits an integer varying from 1 to 10 once every 6 seconds. Board B receives it and shows an LED blinking pattern based on the number received from board A.	(a) Board A takes 6 seconds between transmission of subsequent numbers. (b) When Board A transmits the number '4', the LED started blinking 4 times i.e., number of glows: 4 and duration of each glow is 5000 / (2 * 4) = 625ms, and the total time for all the glows is 5 seconds.

Results for individually assigned task:

Task: 1

[**Video of the result of the task 1**]